to understand this theory, let us first discuss an imaginary experiment. Let us imagine that we have captured an alpha particle which has just emerged from the nucleus and have restored it to the place from which it came, and investigate what forces are acting upon this alpha particle, and what work must be applied. The conditions which we encounter are similar to those which prevail in the proton-neutron relationship (Figure 15). As long as the alpha particle is at a considerable distance from the nucleus, it is subject solely and only to the action of the field, by the positive charge of which it is repelled.

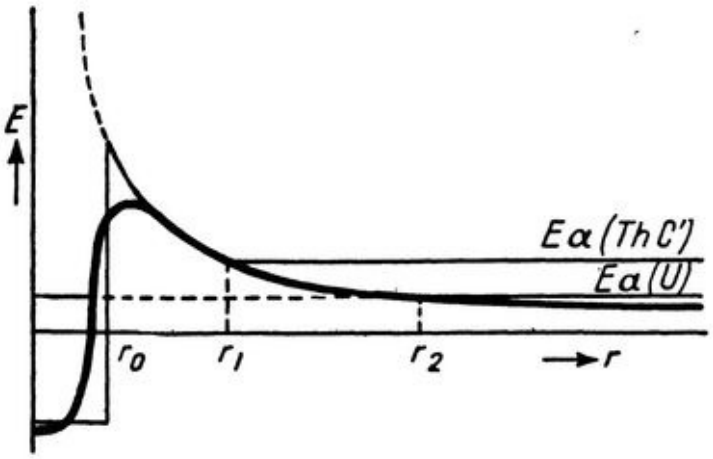


Figure 21.—Potential between heavy nucleus and alpha particle.

Therefore, work has to be applied to the particle in order to bring it closer to the nucleus. This means that as our particle approaches the nucleus, its potential energy increases at first. But when it has come sufficiently close to the nucleus, the short-range nuclear forces of attraction become operative, and they finally overcome the electric repulsion. After a certain potential barrier has been overcome, the force of repulsion changes into a force of attraction, and from there on, the potential energy drops steeply in the direction of the interior of the nucleus. This change in potential energy is shown by the curve in Figure 21.

When the alpha particle is hurled out of the nucleus, it also passes through this potential range, in this case proceeding from inside outwards. Since it reaches a considerable distance from the nucleus with a considerable quantity of kinetic energy, its total energy at that point is a positive magnitude, since its potential energy vanishes. This fact is indicated in Figure 21 by two straight horizontal lines, one of which corresponds to the slow alpha particles of long-lived uranium, the other to the faster alpha particles of the vastly shorter-lived thorium C'. Since the alpha particle carries this energy along out of the nucleus, it must have been in possession of it while still in there. Therefore, we extend the straight line of the uranium right into the interior of